

Basic course

Good to know!

KNX Association

Table of Contents

1	<i>Topology in installations before 2019</i>	4
2	<i>Telegram</i>	5
2.1	General	5
2.2	Telegram structure	5
3	<i>IP Communication</i>	6
3.1	KNXnet/IP Tunneling.....	6
3.2	KNXnet/IP Routing	6
3.3	Difference between a KNX IP interface and a KNX IP Router	7
4	<i>Distributed power supply</i>	8

Preface

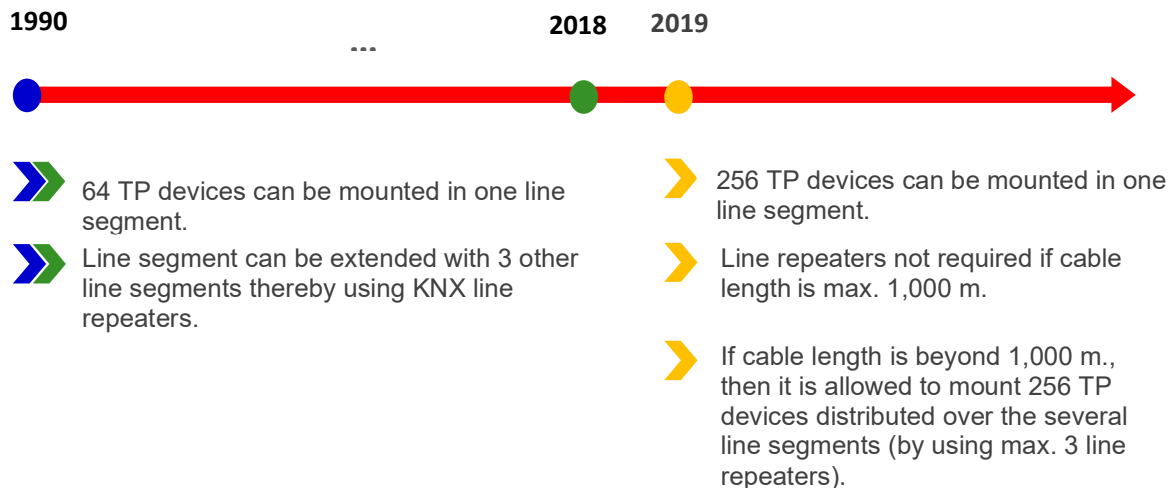
This training document is created for the Basic course. Sections with a heading marked **[E]** contain important information that is also required for the NextGen course. This includes any sub-sections within them. If a heading does **not** have **[E]**, it means the content is only for reference and is not mandatory for the NextGen course.

The documentation focuses on the most used configuration mode in KNX, i.e. S-Mode (System Mode). Configuration modes that used to be supported in the past like E-Mode (Easy Mode) are not covered further.

NOTE: THIS CHAPTER IS INTENDED TO BE USED AS AN INFORMATIVE ANNEX TO THE BASIC COURSE. THE CONTENTS OF THIS CHAPTER IS NOT PART OF THE BASIC COURSE EXAM.

1 Topology in installations before 2019

A distinction shall be made between KNX installations that were planned and commissioned up to 2018 and those that were planned and commissioned later, i.e. from 2019 onwards.



Until 2018 (including):

Only 64 TP devices can be mounted in one line segment. A segment can however be extended by 3 other line segments thereby using 3 line repeaters so that a line consists of up to 4 line segments.

After that:

In newer installations, it is also allowed to mount 256 TP devices in one line without the use of line repeaters. This situation is explained in detail in the Topology chapter.

Important notes:

- The topology of installations that were erected before 2019 **shall be left untouched** and no additional devices beyond 64 shall be installed in one line segment.
- Newer devices are regarded to be “TP-256” compliant, meaning that they allow 256 TP devices to be mounted in one line segment. Older devices are not necessarily TP256 compliant. This explains the reason for distinguishing older installations from newer installations. It is not possible to read from the label whether the device is TP64 or TP256 compliant.

2 Telegram

2.1 General

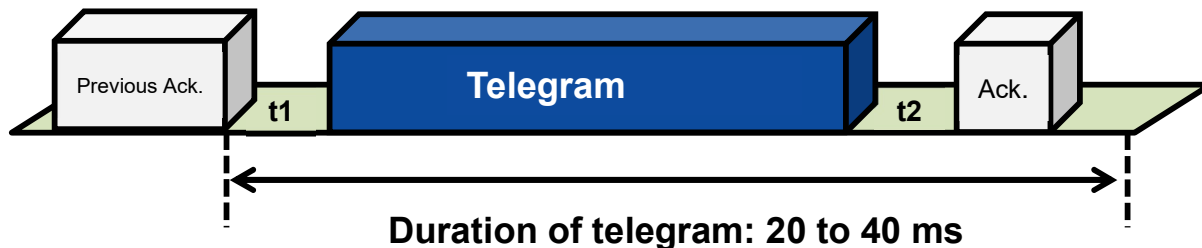


Figure 1: TP telegram: general

When an event occurs, (e.g., when a push button is pressed or a motion is detected etc.), the bus device will send a telegram on the bus. As each device expects after sending a telegram, to receive an acknowledgement, this telegram transmission will start right after the bus has remained unoccupied for at least the time period t1. Once the transmission of the telegram is complete, the bus device will use the time t2 to check whether the telegram has been received correctly. Please note that all addressed bus devices acknowledge the receipt of the telegram simultaneously.

The telegram is transmitted at a bit rate of 9600 bits/sec. This means that one bit occupies the bus for 104 μ s (1/9600 sec). A switch telegram (including acknowledgement) occupies the bus for about 20 milliseconds. Telegrams for text transmission occupy the bus for up to 40 milliseconds.

2.2 Telegram structure

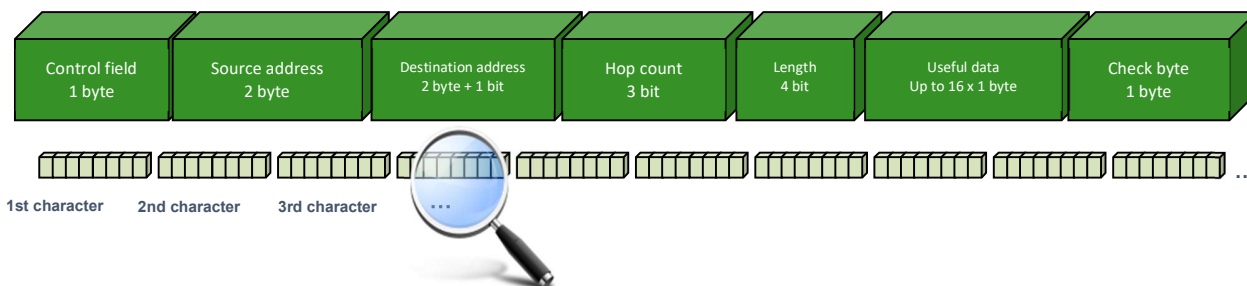


Figure 2: TP telegram structure

The telegram consists of bus-specific data and the actual useful data (= payload), which provides information about the event (e.g., pressing a push button, detecting a movement etc.). The entire information is transmitted in the form of 8-bit long characters.

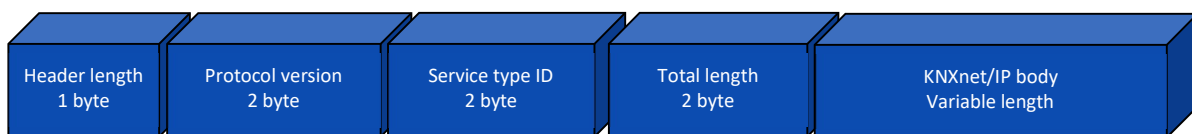
A check byte for the detection of transmission errors is also included in the telegram. This guarantees an extremely high level of transmission reliability when sending telegrams on the bus.

More in-depth information on the TP telegram can be found in the chapter “Telegram” of the KNX Advanced course.

3 IP Communication

KNX supports several media such as TP, RF and IP. The TP cable has the highest level of transmission reliability but has a slower bit transfer rate, i.e. 9600 bits/second compared to IP which can be transferred at a bit rate of 100,000,000 bits/second (100 Mbps).

If there is a visualization running on a TP main or backbone line, this will result in a higher telegram load because of the many status feedback telegrams on the TP primary side. The best option to overcome this load problem is to replace the TP couplers with a KNXnet/IP router as the primary side is IP and is capable of smoothly handling the many status feedback telegrams. The KNXnet/IP telegram on the primary side contains additional information compared to the TP



telegram.

Figure 3: Additional information KNXnet/IP telegram

The field Service type ID indicates which action should be carried out. The two most important services are:

- KNXnet/IP Tunneling
- KNXnet/IP Routing

3.1 KNXnet/IP Tunneling

Tunnelling is necessary for the point-to-point communication of KNX telegrams in an IP frame with ETS. This is always the case if an individual address should be addressed as destination address i.e., when programming the individual address or downloading the application program of a KNX device.

Example: KNX IP Interface, KNX IP Router.

3.2 KNXnet/IP Routing

In KNXnet/IP Routing, a multicast telegram is used for the transmission of KNX telegrams from TP to IP and vice versa. The primary side is IP, which allows a much higher bit transfer rate as explained earlier.

Multicast is like group communication on TP. A KNXnet/IP router wanting to send a line-crossing telegram, will send this telegram with a Multicast IP address onto Ethernet. As all other KNXnet/IP routers are connected to this multicast address they can receive and evaluate this telegram. In this way all telegrams can be seen by all KNXnet/IP devices scanning for this address. KNX Association has reserved the multicast address 224.0.23.12, but any address could be used as long as it is the same on all devices.

Example: KNX IP Router

3.3 Difference between a KNX IP interface and a KNX IP Router

Using a KNX IP Interface or a KNX IP Router is the easiest way to connect the KNX bus to a PC as a USB interface requires a physical cable to connect to the PC to program the KNX devices on the bus. KNX IP interfaces and routers look similar, but the functionality is not the same.

A **KNX IP Interface** enables you to connect to and program the KNX devices on the bus over IP/LAN. It supports KNXnet/IP Tunneling only. Some KNXnet/IP Interfaces may support multiple connections. This is useful when you have multiple instances of ETS accessing the same KNX system.

A **KNX IP Router** has the same functions as an IP Interface but also has in addition to the tunneling function, the functions of a line coupler. The KNX IP router can thus send telegrams in the network to other lines or areas, as well as receive telegrams from these networks. It is possible to configure the filter table to manage the flow of traffic between the primary side and the secondary side just like in a normal TP line coupler.

KNXnet/IP routers can support one or multiple KNXnet/IP Tunneling connections.

More in-depth information on KNX IP can be found in the chapter “IP Communication” of the KNX Advanced course.

4 Distributed power supply

Cable length	Type of bus power supply			
	Decentralized bus power supply with number of DPSU power feeding devices			Central bus power supply unit (PSU)
	1	2	3 ... 8	
Max. total cable length	350 m	700 m	1,000 m	1,000 m
Max. distance between two bus devices	350 m	700 m	700 m	700 m
Max. distance between non-power feeding device and power supply	350 m	350 m	350 m	350 m
Min. distance between two power supply units	No minimal cable distance between two DPSU or between DPSU and standard central PSU			As specified by the manufacturer

Figure 4: Cable lengths when using decentralized power supply units and a centralized power supply unit

In this case, instead of a centralized bus power supply, the bus is powered in a **distributed** way by some devices connected to the line containing each of them a Decentralized Power Supply Unit (DPSU) with integrated choke module. Stand-alone DPSU (non-communicating) devices are also possible.

A DPSU is especially intended for small installations with few devices. Different types of DPSU exist, depending on the output current (25, 40 and 80 mA).

In most cases, it is possible to combine DPSU with up to two standard centralized PSUs.

The DPSU can be located at any location in the bus line. There are no limitations concerning minimal distances between two DPSUs or a DPSU and a standardized central PSU.

Up to 8 DPSUs can be mounted in one single bus line (even if consisting of segments). More than eight can have a negative effect on communication. In case of mounting up to 8 DPSUs in a single line together with a central PSU, the maximum resulting short circuit current of these devices (as given in the product data sheet) shall not exceed 3 A.

In some cases it is possible to manually disable the DPSU on the device (e.g. by means of a switch or configuration of a parameter).

The cable length that needs to be observed in conjunction with the use of centralized and decentralized power supply units is given in the above figure.